

AI RECOMMENDATION SYSTEM

Project Report

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Table of Contents

<u>1</u>	Introduction and Problem Description
<u>2</u>	Theoretical Background
<u>3</u>	Proposed AI Approach
<u>4</u>	System Design and Solution Explanation
<u>5</u>	Queries and Outputs
<u>6</u>	References

Introduction and Problem Description

Artificial intelligence tools have become widely used across different fields, yet choosing the right tool can be challenging due to the large number of available options and the differences in their features, skill requirements, and pricing models. This project aims to address this challenge by developing an AI-based recommendation system that helps users identify the most suitable AI tool based on three key criteria: the type of task they want to perform, their skill level, and their available budget.

The importance of this problem lies in the growing reliance on AI technologies in education, business, and everyday tasks. As the number of AI tools continues to increase, users often struggle to determine which tool best fits their needs. A recommendation system like the one implemented in this project simplifies the decision-making process, saves time, and promotes more effective use of AI resources. By guiding users toward appropriate tools, the system supports better learning outcomes, enhances productivity, and contributes to the broader adoption of AI in society.

Theoretical Background

Artificial intelligence relies on several foundational concepts that enable machines to reason, make decisions, and provide intelligent recommendations. One of the key concepts relevant to this project is expert systems, which are AI programs designed to mimic human decision-making by using predefined knowledge and logical rules.

Expert systems typically operate through a structured knowledge base and an inference engine—both of which are reflected in the design of our Prolog-based recommendation system.

Another essential concept is knowledge representation, which refers to the way information, facts, and relationships are stored so that a machine can understand and process them. In this project, knowledge is represented through Prolog facts that describe AI tools, their task types, skill levels, and budget categories. This structured representation allows the system to retrieve and compare tools efficiently.

The system also relies on inference mechanisms, which are methods used by AI systems to draw conclusions from available information. Prolog's rule-based structure enables the system to infer the best tool for a user by matching their inputs with the stored facts. This process demonstrates logical reasoning, where the system evaluates conditions and applies rules to reach a suitable recommendation.

Theoretical Background

Together, these AI concepts—expert systems, knowledge representation, inference mechanisms, and logical reasoning—form the theoretical foundation of the proposed solution. They enable the system to behave intelligently, analyze user criteria, and provide accurate and meaningful tool recommendations.

Proposed AI Approach

The proposed solution is based on a knowledge-based expert system, which uses predefined facts and logical rules to generate intelligent recommendations.

This approach is suitable for the problem because the system relies on structured information about AI tools—such as task type, required skill level, and budget—to determine the best match for the user.

In this project, facts are used to store information about each AI tool, including its category, difficulty level, and pricing model. These facts form the knowledge base of the system. Rules are then applied to compare the user's input with the stored facts and identify tools that match the given criteria. Prolog's rule-based structure allows the system to perform logical checks and evaluate multiple conditions efficiently.

The system also uses inference techniques, where Prolog's inference engine processes the facts and rules to reach a conclusion. When the user enters their task type, skill level, and budget, the inference mechanism searches the knowledge base and determines whether an exact match exists. If not, the system applies alternative rules to suggest tools with similar characteristics. This demonstrates how logical reasoning is used to solve the problem and provide meaningful recommendations.

System Design and Solution Explanation

The proposed system is designed as a rule-based expert system built using Prolog. Its architecture consists of three main components: a knowledge base, a set of logical rules, and an inference engine. The knowledge base stores all predefined facts about AI tools, including their task categories, required skill levels, and budget types. The rule component defines how the system compares user inputs with the stored facts to determine suitable recommendations. The inference engine processes these rules and facts to generate decisions automatically.

When the user interacts with the system, they provide three inputs: the task type they want to perform, their skill level, and their available budget. Once these inputs are entered, the system begins processing them by searching the knowledge base for tools that match the given criteria. If an exact match is found, the system recommends the corresponding tool. If no perfect match exists, the inference engine applies alternative rules to identify tools that share similar characteristics, ensuring that the user still receives helpful suggestions. This logical flow demonstrates how the system transforms user inputs into meaningful recommendations through structured reasoning.

Queries and Outputs

1. Suggest one perfect match tool based on the user's input (task, level, budget).

`suggest(Tool).`

What is your task type?:
`text`

What is your skill level?:
`advanced`

What is your budget? (free/freemium/paid):
`freemium`

Recommended tool: Claude

Tool	
'Claude'	1

`suggest(Tool).`

What is your task type?:
`images`

What is your skill level?:
`beginner`

What is your budget? (free/freemium/paid):
`free`

No tools match your criteria

Tool	
------	--

true

?- `suggest(Tool).`

2. Suggest one or more perfect match tool based on the user's input (task, level, budget).

`find_by_criteria.`

Enter task:
`images`

Enter your skill level:
`beginner`

Enter budget:
`freemium`

Matching tool(s): [DALL-E, Canva AI]

true	1
------	---

?- `find_by_criteria.`

3. show to the user free tools based on the task they entered.

`show_free_tools.`

Enter task type:
`ml`

FREE TOOLS for ml:

- HuggingFace Models
- TensorFlow
- Scikit-learn

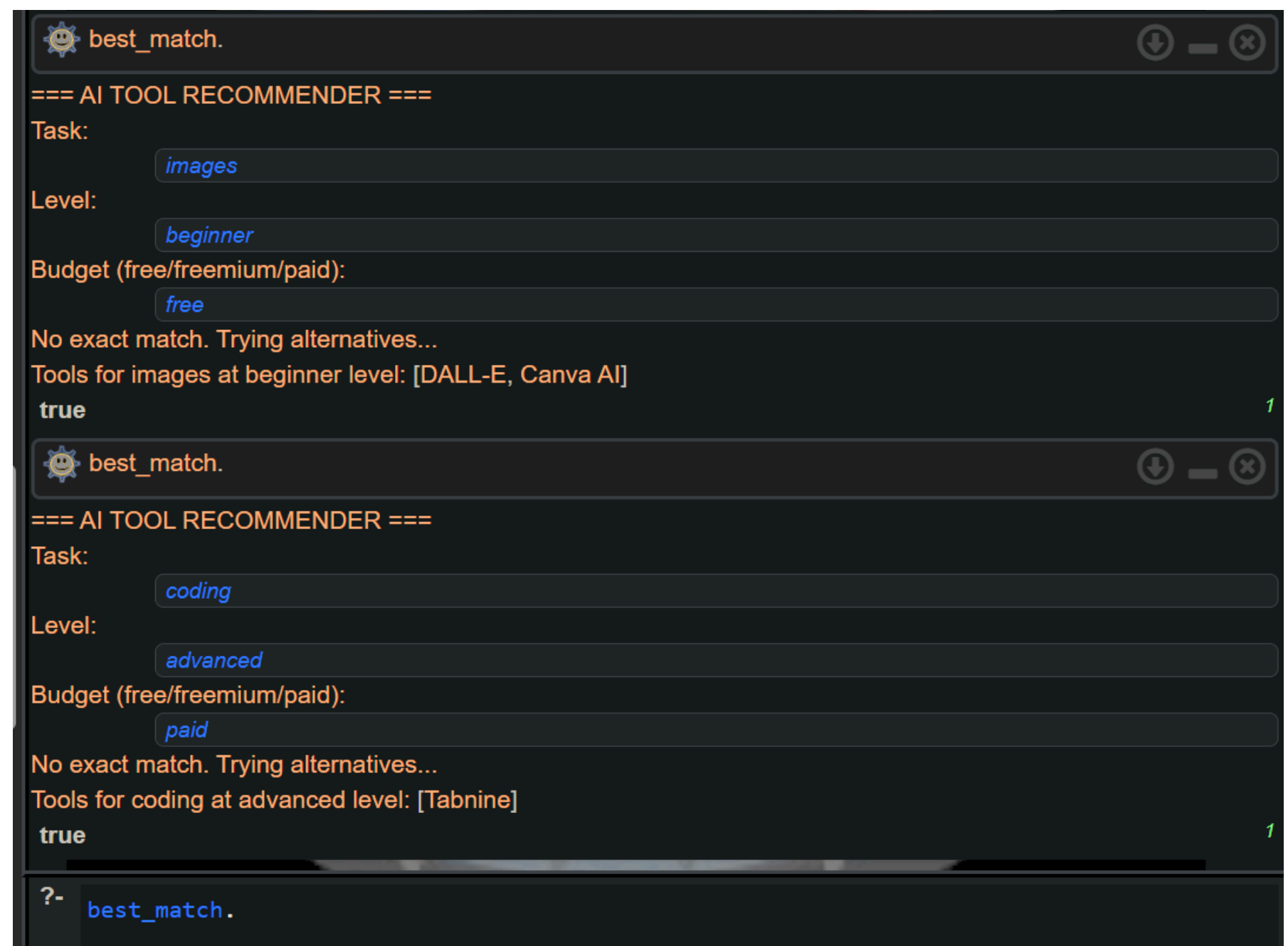
true

true	1
------	---

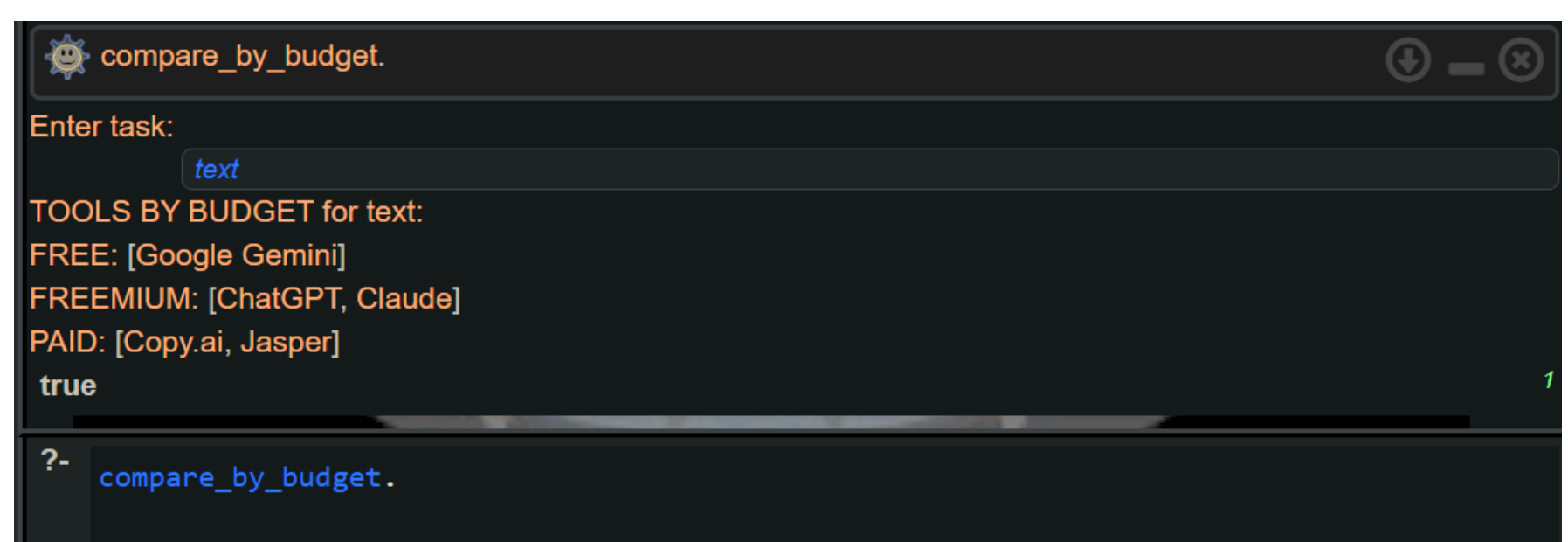
?- `show_free_tools.`

Queries and Outputs

4. Suggest one or more perfect match tool based on the user's input (task, level, budget).



5. User enters a task type, and the system displays all available tools for this task grouped by budget category (free, freemium, paid).



Queries and Outputs

6. Displays all the tools in the database with their full criteria (task, level, and budget).

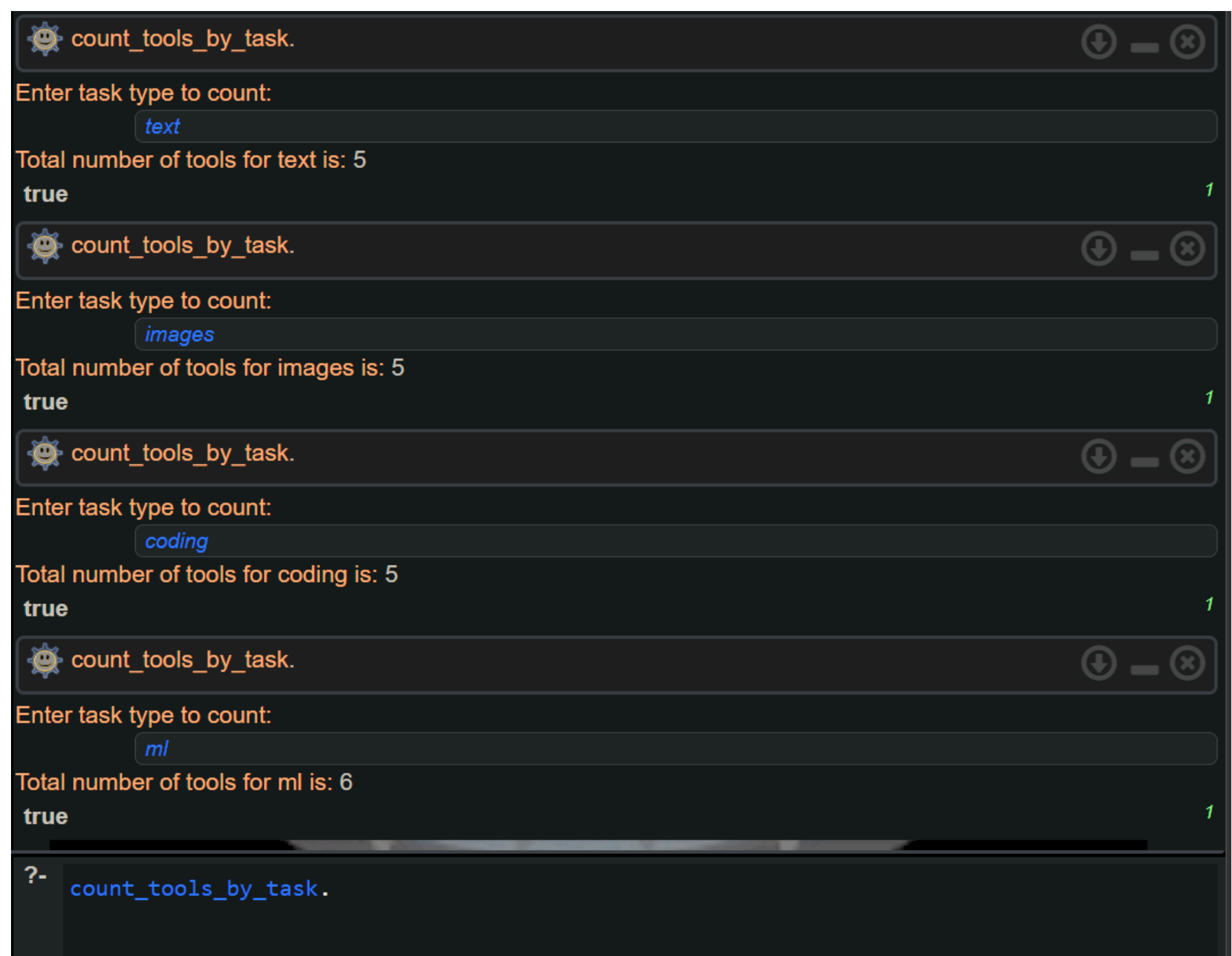
```
list_all_tools.  
  
ALL AI TOOLS:  
- ChatGPT (Task: text, Level: beginner, Budget: freemium)  
- Claude (Task: text, Level: advanced, Budget: freemium)  
- Google Gemini (Task: text, Level: intermediate, Budget: free)  
- Copy.ai (Task: text, Level: beginner, Budget: paid)  
- Jasper (Task: text, Level: intermediate, Budget: paid)  
- Midjourney (Task: images, Level: advanced, Budget: paid)  
- DALL-E (Task: images, Level: beginner, Budget: freemium)  
- Stable Diffusion (Task: images, Level: intermediate, Budget: free)  
- Adobe Firefly (Task: images, Level: intermediate, Budget: paid)  
- Canva AI (Task: images, Level: beginner, Budget: freemium)  
- GitHub Copilot (Task: coding, Level: beginner, Budget: paid)  
- Codeium (Task: coding, Level: intermediate, Budget: free)  
- Tabnine (Task: coding, Level: advanced, Budget: freemium)  
- Amazon CodeWhisperer (Task: coding, Level: beginner, Budget: free)  
- Replit Chostwriter (Task: coding, Level: intermediate, Budget: freemium)  
- HuggingFace Models (Task: ml, Level: advanced, Budget: free)  
- Azure AI Services (Task: ml, Level: intermediate, Budget: paid)  
- Google Cloud AI (Task: ml, Level: advanced, Budget: paid)  
- TensorFlow (Task: ml, Level: intermediate, Budget: free)  
- Scikit-learn (Task: ml, Level: beginner, Budget: free)  
- DataRobot (Task: ml, Level: advanced, Budget: paid)  
true  
?- list_all_tools.
```

7. Show to the user affordable AI tools (Free and Freemium) based on the task they entered by excluding paid options.

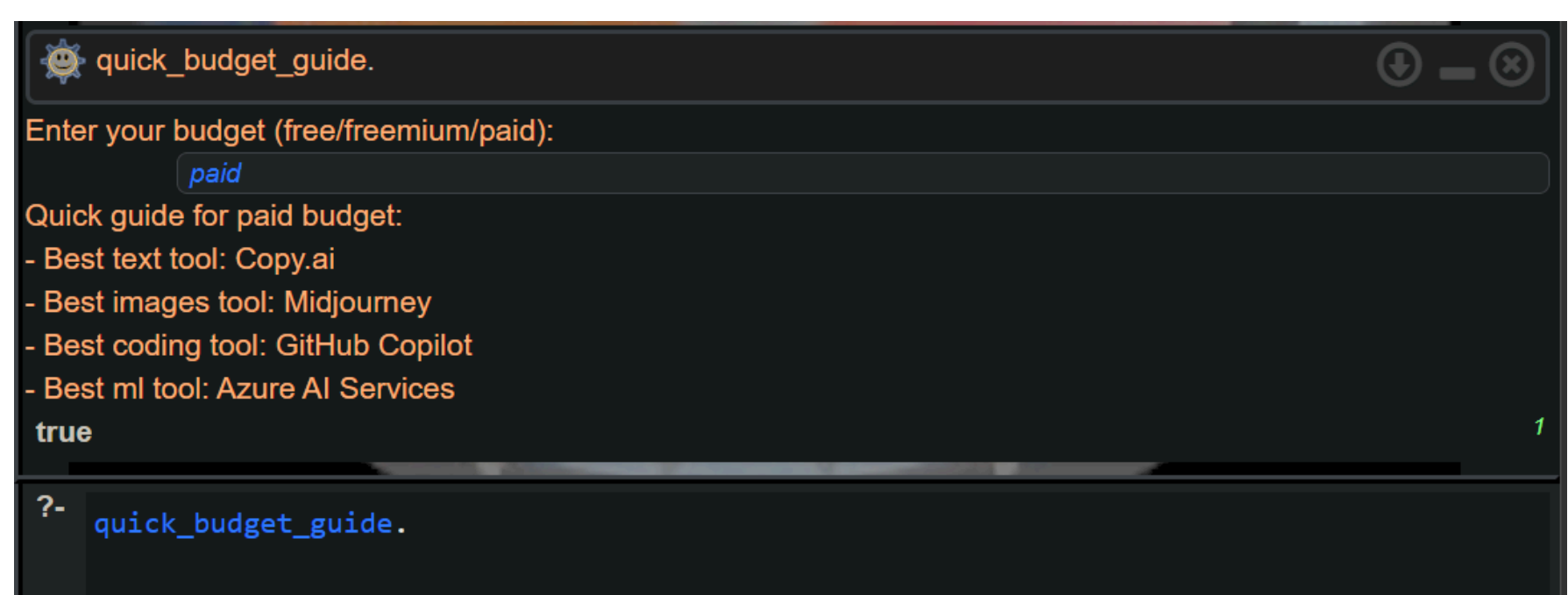
```
show_affordable_tools.  
  
Enter task type to see Free/Freemium options:  
coding  
  
Affordable options for coding:  
- Codeium (free)  
- Tabnine (freemium)  
- Amazon CodeWhisperer (free)  
- Replit Chostwriter (freemium)  
true  
?- show_affordable_tools.
```

Queries and Outputs

- Provides a statistical summary by counting and displaying the total number of AI tools available for a selected task.



- Provides a statistical summary by counting and displaying the total number of AI tools available for a selected task.



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